

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 1 of 10

(1) Research Program/Project/Study Title: Effect of coconut oil as replacement to fish oil in the growth performance of African Catfish, Clarias gariepinus	
Proponent Name:	Rey dela Calzada
Academic Rank:	Associate Professor IV
Co-Authors (with academic rank):	Gelene Roda Alvarez (Instructor I); Robert Christian Patani (Instructor I); Ianthe Marie P. Benliro (Assoc Prof III); Alessandra Nichole Falalimpa (Contractual Faculty)
Campus/Satellite College/Address:	Pontevedra
Telephone/Cell Phone/Fax Number:	09171104034
E-mail Address:	rjdelacalzada@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
--	--

(4) Site of implementation			
		Municipality/City:	Pontvedra
		Province:	Capiz
		Region:	VI
(5) Classification		(6) Mode of Implementation	
Research:	Development:		
☒ Basic	☐ Pilot Testing	☐ Single Authorship/Proponent	
☐ Applied	☐ Tech. Promotion/Commercialization	☒ Collaborative Research (within campus)	
		☐ Collaborative Research (intercampus)	
		☐ Collaborative Research (external agency)	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 2 of 10

(7) Sector/Commodity Aquaculture/Catfish	(8) Discipline/Field of Specialization Fisheries
---	--

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Aquaculture depends primarily on using nutritionally optimized aquafeed that meet the requirements of aquatic organisms for growth and development (Dawood, 2021). Fishmeal and fish oil are the main protei and lipid source in aquaculture feeds (Hodar et al., 2020). Fish meal and fish oils are basically the essential protein and lipid sources associated with their content of essential amino and fatty acids (Kok et al., 2020). Fish meal and fish oils traditionally comes from wild fish sources. In 2018, out of 179 million tonnes of total fish production, about 12% of total catches (22 million tonnes) was reduced to fishmeal and fish oil (FAO, 2020). Fish oil (FO) is a primary lipid used in aquafeed. The major characteristic of FO is the high level of n-3 highly unsaturated fatty acids (HUFA), which are known to be essential for the optimal growth and health of farmed fish (Turchini et al., 2009) and human nutrition (Babalola & Apata, 2011). The increased in aquaculture production in recent years (Tacon, 2020) made these ingredients become more expensive and limited (Turchini et al., 2009) which opened the door to use non-traditional alternatives such as plant protein and lipid sources (Gasco et al., 2018).

Vegetable oils are considered the most sustainable alternatives for fish oil replacement in aquafeeds due to the stable increasing production, high availability and better economic value (Fountoulaki *et al*, 2009). Use of vegetable oils as source of lipids is probed to be successful in fish (Nasopoulou & Zabetakis, 2012; Turchini et al., 2009). Coconut oil is considered to be an efficient and sufficient source of dietary lipid in the aquaculture sector due to its possession of abundant saturated fatty acids (SFAs) (Apraku et al., 2017). Successful incorporation of coconut oil as plant oil source in the diet has been reported in several species of fish such as rainbow trout (*Oncorhynchus mykiss*) (Luo et al., 2014), Nile tilapia (*Oreochromis niloticus*) (Dawood et al., 2021), largemouth bass (*Micropterus salmoides*) (Liang et al., 2022) (large yellow croaker (*Larimichthys crocea*) (Ding et al., 2020), and orange-spotted grouper (*Epinephelus coioides*) (Tseng & Lin, 2020).

Currently, the catfish aquaculture industry in Capiz is one of the top producing industries with an annual growth rate of 5.66% (PSA, 2021). The industry, however, is still dependent on “jako” or trashfish as feed. This aquaculture practice is unsustainable as supply of trashfish is unreliable. Besides, trasfish does not provide the optimum nutrional balance in terms of protein and lipid requirement of the species. The use of coconut oil as an alternative lipid source for fish oil for local aquaculture industry specifically for catfish is encouraging. The use of locally available lipid source (coconut oil) would help boost the local coconut industry. The findings of this research would partly contribute to the formulation of catfish diet utilizing local available sources to help sustain the local catfish aquaculture industry. The output of this research will significantly contribute to human food security and towards the achievement of the United Nations' SDG 2 - Zero Hunger.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The project seeks to determine if coconut oil could be utilized as alternative lipid source to fish oil in catfish aquaculture, in light of the pressing concern of fish oil's increasing demand and becoming more expensive as aquaculture production has tremendously increased in recent years. Locally, catfish

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 3 of 10

production has been on steadily rise but the practice is still traditional and heavily relies on thrashfish (jako) as primary feed. This practice is unsustainable due to unpredictable supply of trashfish from capture fisheries. And, does not supply the optimum nutrients for optimum growth of catfish. The project will focus on finding the best replacement level of coconut oil for fish oil in catfish diet. The replacement levels will be 25%, 50%, 75% and 100%. Juvenile catfish will be distributed randomly to these treatments which will be fed in three replicate hapa nets for 12 weeks. During and at the end of the feeding trial , response parameters such as final average body weight, mean weight gain, specific growth rate, feed intake, feed conversion ratio and survival rate. The project will compliment the proposed project of Mr. Robert Patani on Selected Fermented Plant-based Protein Meal as Replacement to Soybean Meal in the Growth and Survival of Nile Tilapia and Catfish.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Generally, this project proposal will evaluate the effect of coconut oil as alternative to fish oil in *C. gariepinus* growth and survival.

Specifically, this project proposal will evaluate the effect of coconut oil inclusion in the diet in terms of final average body weight, mean weight gain, specific growth rate, feed intake, feed conversion ratio and survival rate.

(11) Review of Literature/Prior Art Search (Related researches/activities that have been conducted. State-of-the-art of current technology/information from which the project proposal will take-off).

Dietary lipids are the sole provider of essential fatty acids and also offers highly digestible energy while enabling the absorption of fat-soluble nutrients essential for proper functioning of physiological processes and in some case maintaining biological structure (Sargent et al., 2003). In Aquaculture, fish oil and fish meal is considered as the main protein components of feed. As the world food fish production has amplified significantly in the recent years, the utilization of fish oil and fish meal is expected to continue to rise in the future. This puts massive pressure on the demand for fish meal (FM) and fish oil (FO) for inclusion in aquafeeds with the consequent increase in the price of these commodities. Thus reducing aquafeeds dependency on FM and FO is of utmost importance and a priority for the sustainable growth of aquaculture (Oliva-Teles et al., 2022).

The most widely used alternatives for fish oil are from vegetable, as production of vegetable oils (VO), is more stable and prices are lower than those for FO (Turchini et al., 2019). According to Nasopoulou et al. (2012), lipids from vegetable sources are usually rich in C18 fatty acids, mainly linoleic (LA, 18:2n-6), -linolenic (ALA, 18:3n-3), and oleic (OA, 18:1n-9), but they lack or have a very limited content of the n-3 LC-PUFA, such as EPA (20:5n-3) and DHA (22:6n-3). The use of vegetable oils as fish oil replacement has been reported to have successfully incorporated in aqua-feed for various fish species with similar or improved growth results as fish oil (Hodar et al., 2020).

Coconut palm, (*Cocos nucifera*) is one of the most common and important plants as its oil is ranked globally, found in almost all part of the world and accounts for 20% of all vegetable oils used worldwide (Aderolu and Akinremi, 2009). Coconut fatty acids composition is: C8:0 (8%), capric acid [C10:0 (7%)], lauric acid [C12:0 (49%)], myristic acid [C14:0 (18%)], palmitic acid [C16:0 (8%)], stearic acid [C18:0 (2%)], oleic acid [C18:1 (6%)], and linoleic acid [C18:2 (2%)] (Yong et al., 2009). This abundant medium chain fatty acids (MCFA, 65%) of the coconut is directly absorbed from the intestine to the liver and is metabolized swiftly for energy production (Dayrit, 2015). The use of lipid in coconut oil as fish oil replacement has been reported in Nile tilapia (Dawood et al., 2021), large yellow croaker (Ding et al., 2020), orange-spotted grouper (Tseng et al., 2020), African catfish (Aderolou & Akinremi, (2009), large rainbow trout (Ballestrazzi & Maxia, 2006) and Pacific white shrimp (Lim et al., 1997).

Dawood et al. (2021) evaluated the influence of coconut oil on the growth, feed utilization, immune, and antioxidative responses of Nile tilapia. Five test diets were formulated by mixing coconut oil with the other ingredients at 0, 1, 2, 3, and 4% of the total ration and presented for tilapia for 60 successive days. The final weight, SGR, weight gain (WG), and feed intake were superior in fish delivered 2% of

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 4 of 10

coconut oil. Concurrently, fish that received 2% coconut oil had lower FCR and higher PER than fish of the control and 4% groups. Higher lipase activity was observed in fish of 2% and 3% levels than the remaining groups. The amylase and protease activities of fish in 1%, 2%, and 3% groups were higher than the 0% level. The total blood cholesterol, RBCs, and PCV showed higher values in Nile tilapia fed 2% and 3% coconut oil. The lysozyme and phagocytic activities were higher in fish fed 2% and 3% levels than the control, while the phagocytic index in 2% and 3% levels was higher than 0% and 4% levels. SOD and CAT were higher in fish fed 1%, 2%, and 3% than fish fed 0% and 4% levels while GSH was higher in fish of 1%, 2%, and 3% than fish fed 0% level. However, the MDA level was markedly lower in fish fed 25, 3%, and 4% coconut oil than the 0% level. The intestine's histological structure in all groups appeared normal, forming of intestinal villi projecting from the intestinal wall. Also, the structure of the hepatopancreas had a normal architecture in all groups. They concluded that inclusion of coconut oil at 2 to 3% is recommended as a replacer for fish oil in Nile tilapia diets.

Ding et al. (2020) investigated the effect of different dietary coconut oil (CO) levels on growth, antioxidant capacity and lipid metabolism of juvenile large yellow croaker (*Larimichthys crocea*). Five iso-nitrogen (45% crude protein) and iso-lipid (13% crude lipid) experimental diets were prepared by replacing 0% (the control), 25%, 50%, 75% and 100% fish oil with coconut oil. The results showed that dietary CO had no significant effect on survival rate. However, the specific growth rate was increased significantly when compared with the control group when fish were fed the diet with 50% CO. The saturated fatty acids were increased significantly with increasing dietary CO in the liver and muscle, whereas the content of n-3 PUFA was decreased significantly. The highest activities of glutathione peroxidase and superoxide dismutase in the liver were recorded in fish-fed diet with 50% CO; conversely, the content of malondialdehyde was significantly decreased. The mRNA expression of peroxisome proliferator-activated receptor α , carnitine palmitoyl transferase 1 and acyl-CoA oxidase reached the highest levels in fish-fed diet with 50% CO. According to the authors, this indicated that the rapid oxidation reaction of fatty acids to provide energy may be the reason for the rapid growth of large yellow croaker. They concluded that fish-fed diet with 50% CO increased the growth rate and antioxidant capacity. Which indicate that the optimal replacement level of CO to FO in the diet should be 50%.

Tseng and Lin (2020) investigated the effects of coconut oil as a dietary supplement on the growth, lipid metabolism and related gene expressions of juvenile orange-spotted grouper *Epinephelus coioides*. Coconut oil at concentrations of 0, 10, 30 and 50 g/kg was used to replace dietary lipids in a basal diet containing 150 g/kg lipids. The four experimental diets were, respectively, fed to triplicate groups of juvenile groupers (initial weight: 8.53 ± 0.13 g) in a recirculating system for 8 weeks. Fish fed the diet containing 50 g/kg coconut oil exhibited lower weight gain than did fish fed the diet containing 30 g/kg coconut oil; however, no significant differences in weight gain were observed between fish fed diets containing 0 and 10 g/kg coconut oil. Hepatic carnitine palmitoyltransferase-1, fatty acid synthase, fatty acid elongase, fatty acid desaturase and peroxisome proliferator-activated receptor gamma gene expressions were all the highest in fish fed the diet containing 10 g/kg coconut oil. Fish fed the coconut oil-free basal diet demonstrated upregulated gene expression of neuropeptide Y. The results of their study suggested that dietary supplementation with 10 g/kg coconut oil exerted beneficial effects on lipid metabolism by *E. coioides*.

The use of coconut oil as alternative source of energy in the diets of African mudcatfish was conducted by Hamed et al. (2014) for twelve weeks to investigate the effects of feeding substituted palm oil (PMO) with coconut oil (CNO) as an alternative source of energy, on the growth performance and nutrient utilization of *Clarias gariepinus* juveniles. Six (6) isonitrogenous 40% crude protein diets were formulated where CNO replaced PMO at 100%, 80%, 60%, 40%, 20% and 0% inclusions and labelled diets 1,2,3,4,5 and 6 respectively. Juveniles of *C. gariepinus* (40.39 ± 2.98 g) stocked at 20 fish/50Lts plastic tanks in triplicate, were fed twice daily to satiation with weight changes recorded weekly. Samples of experimental diets and fish carcass were analyzed for proximate composition, while growth performance and quality of diets were evaluated. Fish fed diet 4 (containing 60% palm oil, 40% coconut oil) inclusion had the highest mean weight 203.36 ± 2.27 g and the best food conversion ratio of 1.44 ± 0.34 g after the twelve weeks of study. There was significant differences in the percentage weight gain between fish fed diet 1 (333.28 ± 1.26 g) and fish fed diet 4 (401.75 ± 10.9 g). No significant difference

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 5 of 10

($P > 0.05$) was recorded in protein intake (PI) between fish fed diets 2, 3 and 4 while significant difference ($P < 0.05$) was recorded in the protein efficiency ratio of fish fed diet 3 (3.05 ± 0.18) and diet 4 (2.50 ± 0.10). Their study implied that for better growth, a 40% partial replacement of PMO with CNO as dietary lipid in diet of African mud catfish (*Clarias gariepinus*) juveniles.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories, and variables that indicate relevant relationships to be tested.)

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Experimental Design

The experimental setup will be composed of 15 units, 1m x 1m x 1m floating nets made of polyethylene black hapa net in an outdoor concrete tank of 76-ton (5.8m x 9.18 m x 1.43 m) capacity. Each hapa net will be fixed at 0.5m away from the side of the tank and 0.5 meters from adjacent hapa net. The hapa nets setup will be supported by bamboo poles. Flow-through water will be provided for 100 percent water exchange daily. Every hapa net will be provided with artificial aeration. During the experiment water temperature will be monitored daily, ammonia will be monitored once a week.

Experimental Diet

Five isonitrogenous diets (average crude protein 40%) and 11% lipid will be formulated following the ingredients of Obeda et al., 2023, with modification of using fish oil instead of soybean oil as source of lipid (Table 1). The control diet will have fish oil as source of lipid while the diets one to four will be composed of four levels of replacement of fish oil with coconut oil (CO) at 25, 50, 75, and 100%. The formulated diet will be sent to the Southeast Asian Fisheries Development Center Feed Mill for production into pellets of 2mm dimension.

Table 1: Ingredients and proximate composition of experimental diets containing different levels of coconut oil (g/kg).

Ingredients	Fish Oil	CO 25%	CO 50%	CO 75%	CO 100%
Fishmeal 65% Anchovy	485.90	485.90	485.90	485.90	485.90
Soybean meal Full fat	146.90	146.90	146.90	146.90	146.90
Rice Bran	103.70	103.70	103.70	103.70	103.70
Cornstarch	150.00	150.00	150.00	150.00	150.00
Fish Oil	30.00	22.50	15.00	7.50	0
Coconut Oil	0	7.50	15.00	22.50	30.00
Vitamin premix	35.00	35.00	35.00	35.00	35.00
CMC	48.50	48.50	48.50	48.50	48.50
Total	1000.00	1000.00	1000.00	1000.00	1000.00
Proximate composition of the basal diet (g kg ⁻¹ dry weight)					
% DM	90.59	90.59	90.59	90.59	90.59
% Crude Protein	40.00	40.00	40.00	40.00	40.00
% Crude Fat	11.82	11.82	11.82	11.82	11.82
% Crude Fiber	3.04	3.04	3.04	3.04	3.04
%Crude Ash	9.75	9.75	9.75	9.95	9.75

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 6 of 10

Nitrogen Free	35.79	35.79	35.79	35.79	35.79
Gross Energy (MJ/kg)	18.93	18.91	18.89	18.87	18.84

Experimental Fish

Four hundred (400) pieces African (*C. gariepinus*) catfish fingerlings, average weight of 20 grams, will be obtained from a private hatchery in Brgy San Pedro, Pontevedra, Capiz. The fish will be stocked in a 2000-liter circular fiberglass tank upon arrival at CapSu Pontevedra Freshwater Fresh Hatchery Center. All fish will be conditioned for 14 days and will be feed with commercial catfish pellet with 36% crude protein (Tateh Feeds) ad libitum twice a day. At the commencement of feeding trial, *C. gariepinus* will be randomly stock in 15 hapa nets at 20 fish per hapa. Three hapa nets will be randomly assigned to each of the 5 experimental diets. Fish will be hand-fed at 5% body weight twice a day (9:00am and 5:00 pm) over a period of twelve weeks. .

Sampling Procedures and Growth Parameters Evaluation

Fish will be weighed at the beginning of the experiment and weekly throughout 12 weeks of the experimental period. Feed intake will be monitored for each experimental group in order to measure average feed intake and their effects on growth. These will be determined by evaluating a number of growth indexes which include weight gain, specific growth rate (SGR), feed intake, feed conversion ratio (FCR), survival rate.

Data Computation

The following formula will be used in the analysis:

Mean weight gain = mean final weight – mean initial weight

Specific Growth Rate = $\frac{(\text{Log}W_2 - \text{Log}W_1)}{T_2 - T_1} \times 100$
(%/day)

where, W2 and W1 represent – final and initial weight,
T2 and T1 represent – final and initial time

Feed intake (FI) g = Sum of daily feed weight

Feed conversion ratio = $\frac{\text{Feed fed on dry matter}}{\text{Fish live weight gain}}$
(FCR)

Survival rate (SR) = 100(Number at end of feeding trial/Number at start of feeding trial).

Statistical Analysis

Data will be entered using Microsoft Excel. Analysis of variance (ANOVA) will be used to determine the variation among the treatments while Duncan multiple range test will be used to separate the means. All analyses will be performed at 95% confidence interval. Statistical Analysis will be done using Statistical Package for Social Sciences (SPSS) software (Version 22.0 for Windows; SPSS Inc., Chicago, IL, USA).

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 7 of 10

Research Objectives	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr
1. Feeding Trial										
2. Analysis of data gathered from observation										
3. Report writing										

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible.)

The expected output of this project will be the determination of the best replacement level of coconut oil for fish oil, that will provide the best result in the parameters measured. The result of this study will become one of the reference in formulating practical diet for African catfish.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The potential product that will be generated from this project is a formulated diet incorporated with coconut oil for *C. gariepinus*. A phase 2 of the project will be field trial of the diet. A collaboration with local catfish growers association in Brgy San Pedro, Pontevedra will be pursued. This association will be tapped to conduct the field trial in their catfish farms. Capsu-Pontevedra will be responsible for the production of the formulated feeds and provide regular technical assistance during the field trial. The collaborator will be incharged of other inputs (e.g. catfish fingerlings), pond preparation and day-to-day cultural management. Data gathering (biological parameters sampling) shall be done monthly up to harvest. At the end of the production run. Economic aspects shall also be determined. Once the trial is found successful, a long term collaboration with the association shall be pursued for extension activities. The extension program shall include not only the technical aspect of the project but also the financial and organizational management. To capacitate the catfish grower association, training on catfish grow-out culture using formulated diet as feeds, feed formulation, financial illiteracy, entrepreneurial mindset and organizational management will be conducted at the onset of the collaboration.

(16) Targets Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

The target beneficiaries of this project will be the major stakeholders in the catfish industry (growers, operators and feed producers), university researchers, research institutions and government institution.

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Rey J. dela Calzada	Project Leader	Fisheries (Aquaculture)
Robert Christian F. Patani	Project Member	Fisheries (Aquaculture)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 8 of 10

Gelene Roda I. Alvarez	Project Member	Fisheries (Aquaculture)
Ilanthe Marie P Benliro	Project Member	Fisheries (Marine Fisheries)
Alessandra Nicole B. Falalimpa	Project Member	Fisheries (Post Harvest)

Percent Time Devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 2 hours/day

Program/Project Members (in hours): 1 hour/day

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (Php)
A. Personnel Services Research Assistant (@12,000/month x 4 months)	1 (one)	48,000.00
B. Maintenance and Other Operating Expenses (MOOE)		
- Transportation allowance		5,000.00
- Hapa net set-up		25,000.00
- Feed Formulation		40,000.00
- Feed Proximate Analysis		20,000.00
- Office Supplies		3,000.00
- Feeding trial support materials		5,000.00
- Supply water pipes and fittings		15,000.00
- Water Parameters test kit		8,000.00
- Digital weighing balance		8,000.00
- Miscellaneous		5,000.00
C. CAPITAL OUTLAY		
- Refrigerator (5cu.ft.)		15,000.00
Total		197,000.00

(19) Literature Cited (Include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Aderolu, A. Z., & Akinremi, O. A. (2009). Dietary effects of coconut oil and peanut oil in improving biochemical characteristics of *Clarias gariepinus* juvenile. *Turkish Journal of Fisheries and Aquatic Sciences*, 9(1).

Apraku, A., Liu, L., & Ayisi, C. L. (2017). Trends and status of dietary coconut oil in aquaculture feeds. *Reviews in Fisheries Science & Aquaculture*, 25(2), 126-132.

Babalola, T. O. O., & Apata, D. F. (2011). Chemical and quality evaluation of some alternative lipid sources for aqua feed production. *Agriculture and Biology Journal of North America*, 2(6), 935-943.

Ballestrazzi, R., Rainis, S., & Maxia, M. (2006). The replacement of fish oil with refined coconut oil in the diet of large rainbow trout (*Oncorhynchus mykiss*). *Italian Journal of Animal Science*, 5(2), 155-164.

Dawood, M. A. (2021). Nutritional immunity of fish intestines: Important insights for sustainable aquaculture. *Reviews in Aquaculture*, 13(1), 642-663.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 9 of 10

Dawood, M. A., Ali, M. F., Amer, A. A., Gewaily, M. S., Mahmoud, M. M., Alkafafy, M., ... & Van Doan, H. (2021). The influence of coconut oil on the growth, immune, and antioxidative responses and the intestinal digestive enzymes and histomorphometry features of Nile tilapia (*Oreochromis niloticus*). *Fish Physiology and Biochemistry*, 47, 869-880.

Dayrit, F. M. (2015). The properties of lauric acid and their significance in coconut oil. *Journal of the American Oil Chemists' Society*, 92, 1-15.

Ding, T., Xu, N., Liu, Y., Li, X., Xiang, X., Xu, D., ... & Ai, Q. (2020). Optimal amounts of coconut oil in diets improve the growth, antioxidant capacity and lipid metabolism of large yellow croaker (*Larimichthys crocea*). *Marine Life Science & Technology*, 2, 376-385.

FAO. 2020. The State of World Fisheries and Aquaculture 2020. Sustainability in action. Rome. <https://doi.org/10.4060/ca9229en>

Fountoulaki, E., Vasilaki, A., Hurtado, R., Grigorakis, K., Karacostas, I., Nengas, I., ... & Alexis, M. N. (2009). Fish oil substitution by vegetable oils in commercial diets for gilthead sea bream (*Sparus aurata* L.); effects on growth performance, flesh quality and fillet fatty acid profile: Recovery of fatty acid profiles by a fish oil finishing diet under fluctuating water temperatures. *Aquaculture*, 289(3-4), 317-326.

Gasco, L., Gai, F., Maricchiolo, G., Genovese, L., Ragonese, S., Bottari, T., ... & Caruso, G. (2018). Fishmeal alternative protein sources for aquaculture feeds. Feeds for the aquaculture sector: current situation and alternative sources, 1-28.

Hammed, A. M., Amosu, A. O., Fashina-Bombata, H. A., Ajepe, R. G., & Olufowobi, M. T. (2014). The use of coconut (*Cocos nucifera*) oil as an alternative source of energy in the diets of African Mudcatfish (*Clarias gariepinus*—Burchell, 1822) juveniles. *IJSE R*, 5(8), 431-436.

Hodar, A. R., Vasava, R. J., Mahavadiya, D. R., & Joshi, N. H. (2020). Fish meal and fish oil replacement for aqua feed formulation by using alternative sources: a review. *Journal of Experimental Zoology India*, 23(1).

Kok, B., Malcorps, W., Tlusty, M. F., Eltholth, M. M., Auchterlonie, N. A., Little, D. C., ... & Davies, S. J. (2020). Fish as feed: Using economic allocation to quantify the Fish In: Fish Out ratio of major fed aquaculture species. *Aquaculture*, 528, 735474.

Liang, C., Zhao, X., Jiao, L., Shen, Y., Luo, J., Zhu, T., ... & Jin, M. (2022). Effects of different lipid sources on growth performance, fatty acids composition in tissue and expression of genes related to lipid metabolism in largemouth bass (*Micropterus salmoides*). *Aquaculture Reports*, 23, 101013.

Lim, C., Ako, H., Brown, C. L., & Hahn, K. (1997). Growth response and fatty acid composition of juvenile *Penaues vannamei* fed different sources of dietary lipid. *Aquaculture*, 151(1-4), 143-153.

Luo, L., Xue, M., Vachot, C., Geurden, I., & Kaushik, S. (2014). Dietary medium chain fatty acids from coconut oil have little effects on postprandial plasma metabolite profiles in rainbow trout (*Oncorhynchus mykiss*). *Aquaculture*, 420, 24-31.

Nasopoulou, C., & Zabetakis, I. (2012). Benefits of fish oil replacement by plant originated oils in compounded fish feeds. A review. *LWT*, 47(2), 217-224.

Obeda, R. L., Tumbokon, B. L., & Serrano Jr, A. E. (2023). Dietary benfotiamine in high carbohydrate diet improves growth and resistance to abrupt shift to higher salinity in the African catfish *Clarias gariepinus* juveniles. *Aquaculture, Aquarium, Conservation & Legislation*, 16(3), 1266-1276.

	Document Type:	Document Code	RES- RP-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	November 30, 2019
	RESEARCH PROPOSAL	Page	Page 10 of 10

Oliva-Teles, A., Enes, P., Couto, A., & Peres, H. (2022). Replacing fish meal and fish oil in industrial fish feeds. *Feed and Feeding Practices in Aquaculture*, 231-268

PSA. (2021). "Aquaculture: Volume of Production by Species, Geolocation, Year and Quarter."
https://Openstat.psa.gov.ph/PXWeb/Pxweb/En/DB/DB__2E__FS/0092E4GVAP1.Px/?Rxid=bdf9d8da-96f1-4100-ae09-18cb3eaeb313t.

Sargent, J. R., Tocher, D. R., & Bell, J. G. (2003). The lipids. *Fish nutrition*, 181-257.

Tacon, A. G. (2020). Trends in global aquaculture and aquafeed production: 2000–2017. *Reviews in Fisheries Science & Aquaculture*, 28(1), 43-56.

Tseng, Y., & Lin, Y. H. (2020). Effects of dietary supplementation with coconut oil on the growth, fatty acid profiles and some lipid metabolism relative gene expressions of orange-spotted grouper *Epinephelus coioides*. *Aquaculture Nutrition*, 26(1), 201-210.

Turchini, G. M., Torstensen, B. E., & Ng, W. K. (2009). Fish oil replacement in finfish nutrition. *Reviews in Aquaculture*, 1(1), 10-57.

Turchini, G. M., Trushenski, J. T., & Glencross, B. D. (2019). Thoughts for the future of aquaculture nutrition: realigning perspectives to reflect contemporary issues related to judicious use of marine resources in aquafeeds. *North American Journal of Aquaculture*, 81(1), 13-39.

Yong, J. W., Ge, L., Ng, Y. F., & Tan, S. N. (2009). The chemical composition and biological properties of coconut (*Cocos nucifera* L.) water. *Molecules*, 14(12), 5144-5164.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)